

PRITHU VERMA

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Education

SRM Institute of Technology, Delhi NCR

Aug 2023 - July 2027

Bachelor of Technology, Computer Science and Engineering, CGPA: 8.02

Coursework: Database Systems, Data Mining, Machine Learning, Natural Language Processing

Affiliations: SRM ACM, TedxSRM, IEEE Computer Society, SRM GeeksForGeeks

Awards: Smart India Hackathon Runner Up, HackerRank Gold Medal, AIR 99 CodingNinjas Gemini Quest, IELTS Band 7.0

Skills

Languages	Python, SQL, Bash, C/C++, Java, HTML/CSS, JavaScript
Data	PostgreSQL, ETL Pipeline, PySpark, PowerBI, DAX, PowerQuery, MySQL, MongoDB, Databricks
ML/AI	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, DBSCAN
Tools	Docker, Git, GitHub Actions, Weights & Biases, MLflow

Publications

- Verma, P., Verma, A. "A Hybrid Unsupervised Learning Framework for Multi-Type Anomaly Detection in Database Query Logs Accepted" **Under Review IEEE INDICON 2026**
- Verma, A., Iqbal, F., Dutt, R., Verma, P., Ray Choudhury, S. "Multi-Dimensional Evaluation of Large Language Models for Causal Reasoning Tasks" **Under review ACL ARR 2026**

Experience

Undergraduate Researcher – SRM Institute of Technology, Delhi NCR

Jan 2026 – July 2026

- Designed an unsupervised anomaly detection framework using Python, PostgreSQL, and scikit-learn on **10,000** query logs with **725 anomalies** across security, temporal, and performance categories
- Engineered **35 behavioral features** and evaluated Isolation Forest, **DBSCAN**, and **One-Class SVM**, achieving best F1 scores of **0.972** temporal, **0.714** security, and **0.634** performance
- Implemented DBSCAN with One-Class SVM ensemble achieving 1.00 recall across all 725 anomalies with zero false negatives and an overall F1-score of 0.794

Projects

Health Insurance Claim Analysis – Python, SQL, SQLAlchemy, ETL Pipeline

- Processed **750,000+ synthetic healthcare records** using Python, Faker, SQLAlchemy, and an ETL pipeline across **6 normalized MySQL tables**, maintaining referential integrity for large-scale multi-table analysis
- Reduced dataset storage by **60–70%** using Python to export generated data to Parquet, enabling compact intermediate storage and efficient bulk ingestion into MySQL
- Improved data consistency using **primary/foreign keys, schema constraints, and validation checks**, preventing orphaned records and preserving relationships across **6 tables**
- Analyzed insurance claim patterns using **SQL joins, aggregations, and analytical queries**, extracting trends across claim costs, approval rates, providers, and patient utilization

Stock Portfolio – Python, PostgreSQL, ETL Pipeline, Streamlit, yfinance

- Designed a 6-table PostgreSQL schema and defined 15+ advanced SQL queries using CTEs, window functions, joins, and aggregations to analyze holdings, transactions, asset allocation, and portfolio performance
- Built a Python ETL pipeline using yfinance to ingest and process multi-year historical market data, integrating price data with 6 relational tables for automated portfolio analysis
- Calculated 5+ financial KPIs, including returns, cumulative performance, volatility, Sharpe ratio, and position-level P&L, surfacing results through an interactive Streamlit dashboard
- Devised analytical queries for position-level gains/losses, portfolio allocation, historical returns, and security performance, converting normalized transactional data into reusable portfolio KPIs

Business Analysis Dashboards – PowerBI, DAX, PowerQuery

- Developed 3 end-to-end Power BI dashboards across banking, toy retail, and superstore datasets, creating 30+ interactive visualizations to analyze revenue, sales, customers, products, and operational performance
- Created reusable KPIs and calculated measures to summarize business performance across several dimensions, transforming raw transactional datasets into management-level metrics and trend analyses
- Represented slicers, drill-throughs, cross-filtering, and interactive visual relationships across 30+ visuals, enabling users to move from high-level KPIs to detailed analysis without modifying underlying reports